Q1. Explain Performance Improvement in Machine Learning

Performance improvement in machine learning means making a model more accurate and reliable. It is done by using good quality data, selecting important features, choosing the right algorithm, and tuning parameters. Techniques like regularization and cross-validation help avoid errors and improve results.

Key Points:

Use clean and relevant data

Select suitable model and features

Apply tuning and validation techniques

OR

Q2. Difference between Predictive and Descriptive

Predictive Analysis	Descriptive Analysi
Predicts future outcomes from past data.	Describes and summarizes past data.
Focuses on future prediction.	Focuses on past understanding.
Uses machine learning models.	Uses basic statistics and charts.
More complex.	Simpler.
Example: Predicting marks or sales.	Example: Showing past results or reports.

Q3. Define Ensemble for Performance Improvement

Ensemble learning is a technique in machine learning where multiple models are combined to improve overall performance and accuracy. Instead of relying on a single model, several models are trained and their outputs are combined to produce a better result.

The main idea is that different models may make different errors, so combining them reduces the overall error and increases reliability. Common ensemble methods include bagging, boosting, and stacking.

Key Points:

Combines multiple models

Improves accuracy and reduces errors

Examples: Random Forest, Boosting

Q4. Describe Model Representation and Interpretability

Model representation refers to how a machine learning model is structured and how it captures the relationship between input features and output. It defines the form of the model, such as linear models, decision trees, or neural networks, and determines how the model learns patterns from data.

Interpretability refers to how easily a human can understand the decisions or predictions made by the model. A model with high interpretability allows users to know why a particular output was given, while complex models may act like a “black box” and are harder to understand.

Q5. Examples of Supervised Learning

1.Predicting the correct drug dosage for a patient based on medical data (Healthcare).

2.Classifying diseases using symptoms and test results (Healthcare).

3.Email spam detection (IT field).

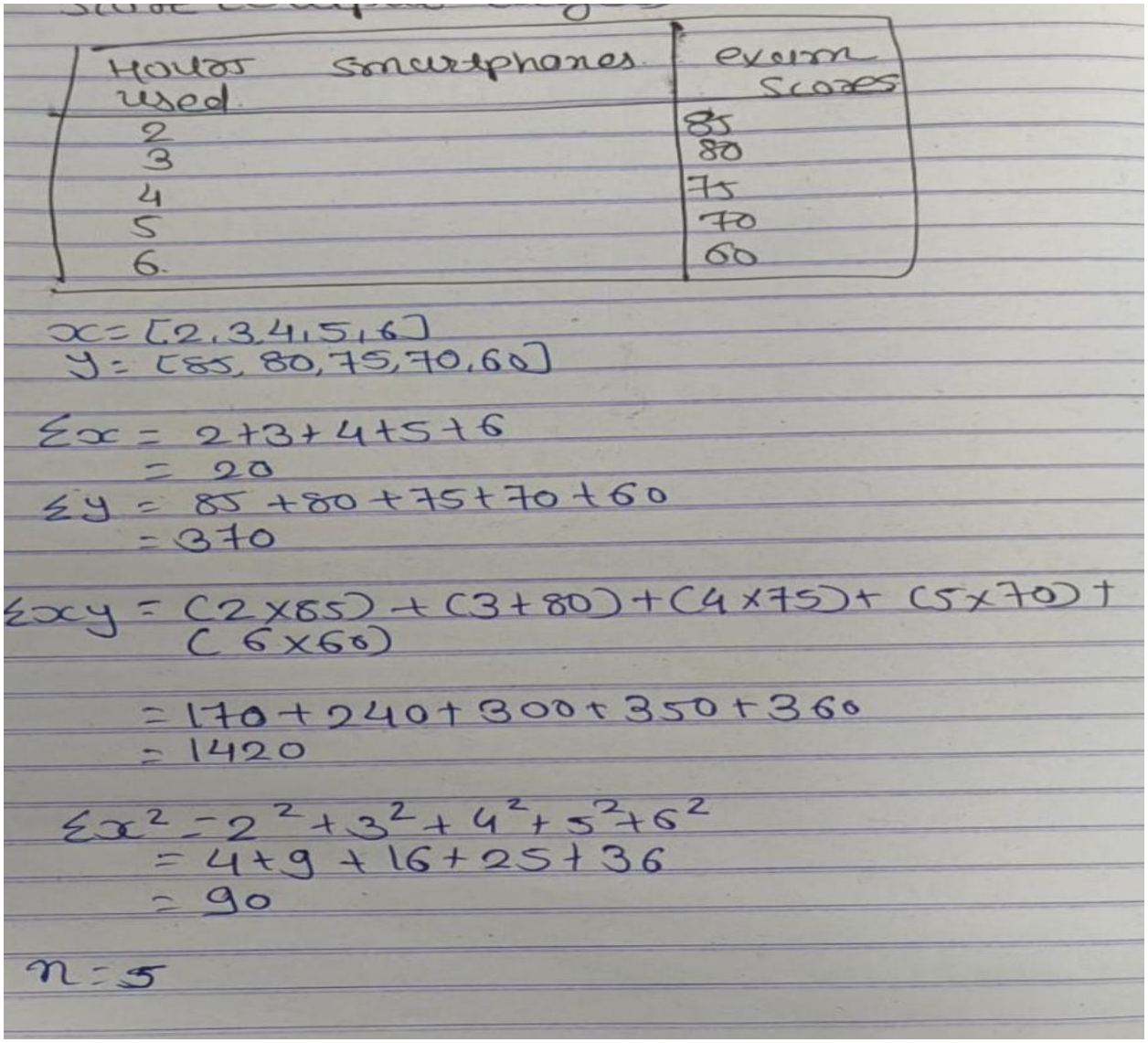
4.House price prediction based on location and size (Real estate).

OR

Q6. Applications of Regression

1. House price prediction based on location and size
2. Sales forecasting in business
3. Student marks prediction based on study hours

Q7. Find a linear relationship between smartphone usage and exam score



Hours used	Smartphones	exam Scores
2		85
3		80
4		75
5		70
6.		60

$x = [2, 3, 4, 5, 6]$

$y = [85, 80, 75, 70, 60]$

$\Sigma x = 2+3+4+5+6$

$= 20$

$\Sigma y = 85+80+75+70+60$

$= 370$

$\Sigma xy = (2 \times 85) + (3+80) + (4 \times 75) + (5 \times 70) + (6 \times 60)$

$= 170 + 240 + 300 + 350 + 360$

$= 1420$

$\Sigma x^2 = 2^2 + 3^2 + 4^2 + 5^2 + 6^2$

$= 4 + 9 + 16 + 25 + 36$

$= 90$

$n = 5$

$$m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

$$m = \frac{5(1420) - (20)(370)}{5(90) - (20)^2}$$

$$m = \frac{7100 - 7400}{450 - 400}$$

$$= \frac{-300}{50}$$

$$= -60$$

Intercept (b)

$$b = \frac{\Sigma y - m\Sigma x}{n}$$

$$b = \frac{370 - (-6 \times 20)}{5}$$

$$b = \frac{370 + 120}{5}$$

$$b = \frac{490}{5}$$

$$= 98$$

$$y = -6x + 98$$

linear relationship

Q8. Explain Support Vectors in the SVM Model

Support vectors are the data points closest to the decision boundary in a Support Vector Machine. They determine the position of the hyperplane and define the margin between classes. Any change in these points directly affects the decision boundary, making them essential for accurate classification. They are selected from the training data during model building. Other data points do not influence the boundary as much as support vectors do.

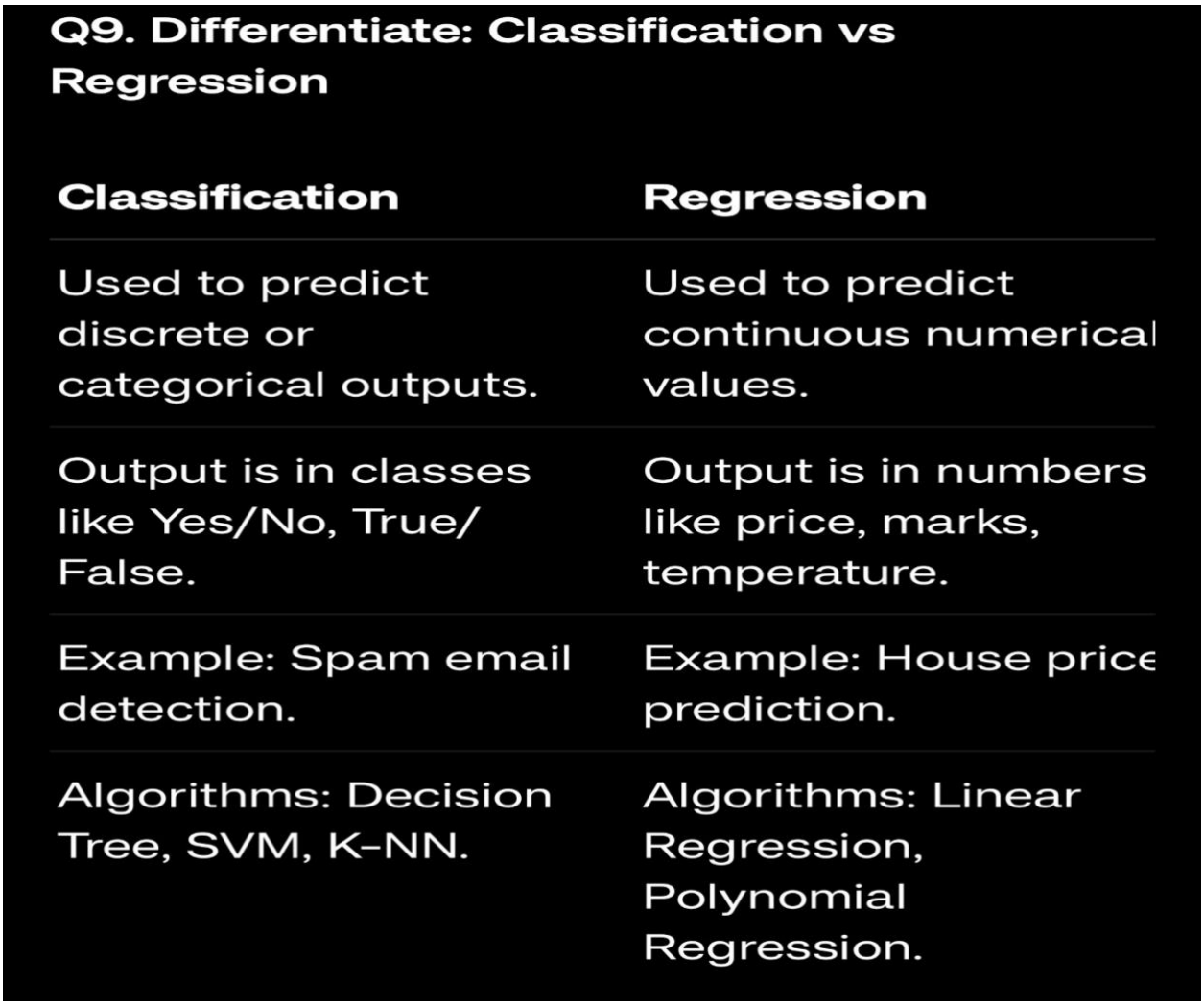
Application of SVM (Support Vectors)

Image classification (face detection, handwritten digit recognition)

Text classification (spam email detection)

Bioinformatics (gene or protein classification)

Q9. Differentiate: Classification vs Regression



Q10. Explain K-NN Algorithm

K-Nearest Neighbors (K-NN) is a supervised learning algorithm that predicts output based on the nearest ‘k’ data points. It calculates distance and uses majority voting for classification or averaging for regression. It is simple but slow for large datasets.

It does not require training phase as it stores all data. The value of ‘k’ affects accuracy of the model. It is sensitive to noise and irrelevant features.

Q11. Explain different types of Logistic Regression in brief

Logistic regression is used for classification problems where the output is categorical.

Binary Logistic Regression

Used when there are only two classes such as Yes/No or 0/1. It predicts the probability of one class using a sigmoid function. It is widely used in spam detection and disease prediction. It is simple and easy to implement.

Multinomial Logistic Regression

Used when there are more than two categories without any order. It uses softmax function to handle multiple classes. It is applied in problems like image classification or object recognition. It extends binary logistic regression to multiple classes.

Ordinal Logistic Regression

Used when categories have a meaningful order like low, medium, high. It considers the ranking between categories during prediction. It is used in rating systems and surveys. It provides better results when order matters.

Q12. Differentiate between Supervised and Unsupervised Learning

Supervised Learning	Unsupervised Learning
Uses labeled data for training. Output is known.	Uses unlabeled data. Output is unknown.
Learns mapping between input and output.	Finds hidden patterns or relationships.
Used for classification and regression.	Used for clustering and association.
Requires training with correct answers.	No predefined answers are given.
More accurate due to labeled data.	Less accurate but useful for discovery.
Algorithms: SVM, Decision Tree, K-NN.	Algorithms: K-means, Apriori.
Example: Spam detection, price prediction.	Example: Customer segmentation, market analysis.

Q13. Differentiate between Clustering vs Classification

Clustering	Classification
Groups data into clusters based on similarity.	Assigns data to predefined classes.
Unsupervised learning technique.	Supervised learning technique.
No labeled data is required.	Requires labeled dat for training.
Output is clusters without predefined labels.	Output is fixed categories or classe
Example: Customer segmentation.	Example: Spam ema detection.

Q14. Applications of Unsupervised Learning

1. Market basket analysis to find items frequently bought together
2. Anomaly detection in fraud detection systems
3. Image segmentation in computer vision
4. Document or text clustering in natural language processing
5. Recommendation systems (grouping users with similar interests)
6. Data compression to reduce size of large datasets

Q15. Explain any two applications of the Apriori Algorithm

1. market basket analysis, where it identifies products that are frequently bought together. This helps stores arrange items and create combo offers to increase sales.

2. recommendation systems, where it suggests products to users based on their previous purchases or behavior. This improves user experience and increases engagement.

Q16. Perform K-means clustering (k = 2) – Step by Step

Given points:

{(2,3), (3,3), (4,3), (5,3), (6,3), (7,3), (8,3), (25,20), (26,20), (27,20), (28,20), (29,20), (30,20)}

Step 1: Select k = 2

Number of clusters = 2

Step 2: Initialize centroids

Assume:

C1 = (2,3)

C2 = (25,20)

Step 3: Assign points to nearest centroid

Points near (2,3) → Cluster 1

Points near (25,20) → Cluster 2

Cluster 1: (2,3), (3,3), (4,3), (5,3), (6,3), (7,3), (8,3)

Cluster 2: (25,20), (26,20), (27,20), (28,20), (29,20), (30,20)

Step 4: Recalculate centroids

C1 = average of Cluster 1 = (5,3)

C2 = average of Cluster 2 = (27.5,20)

Step 5: Reassign points

Check distances again using new centroids.

All points remain in same clusters (no change).

Step 6: Stop condition

No change in clusters → algorithm converged

Final Answer:

Cluster 1 → (2,3) to (8,3), Centroid = (5,3)

Cluster 2 → (25,20) to (30,20), Centroid = (27.5,20)

Q17. Describe the K-means clustering algorithm with a real-world example and diagram

K-means clustering is an unsupervised learning algorithm used to group data into k clusters based on similarity. Each cluster is represented by a centroid, and data points are assigned to the nearest centroid.

Algorithm (Steps):

Select number of clusters (k)

Initialize k centroids randomly

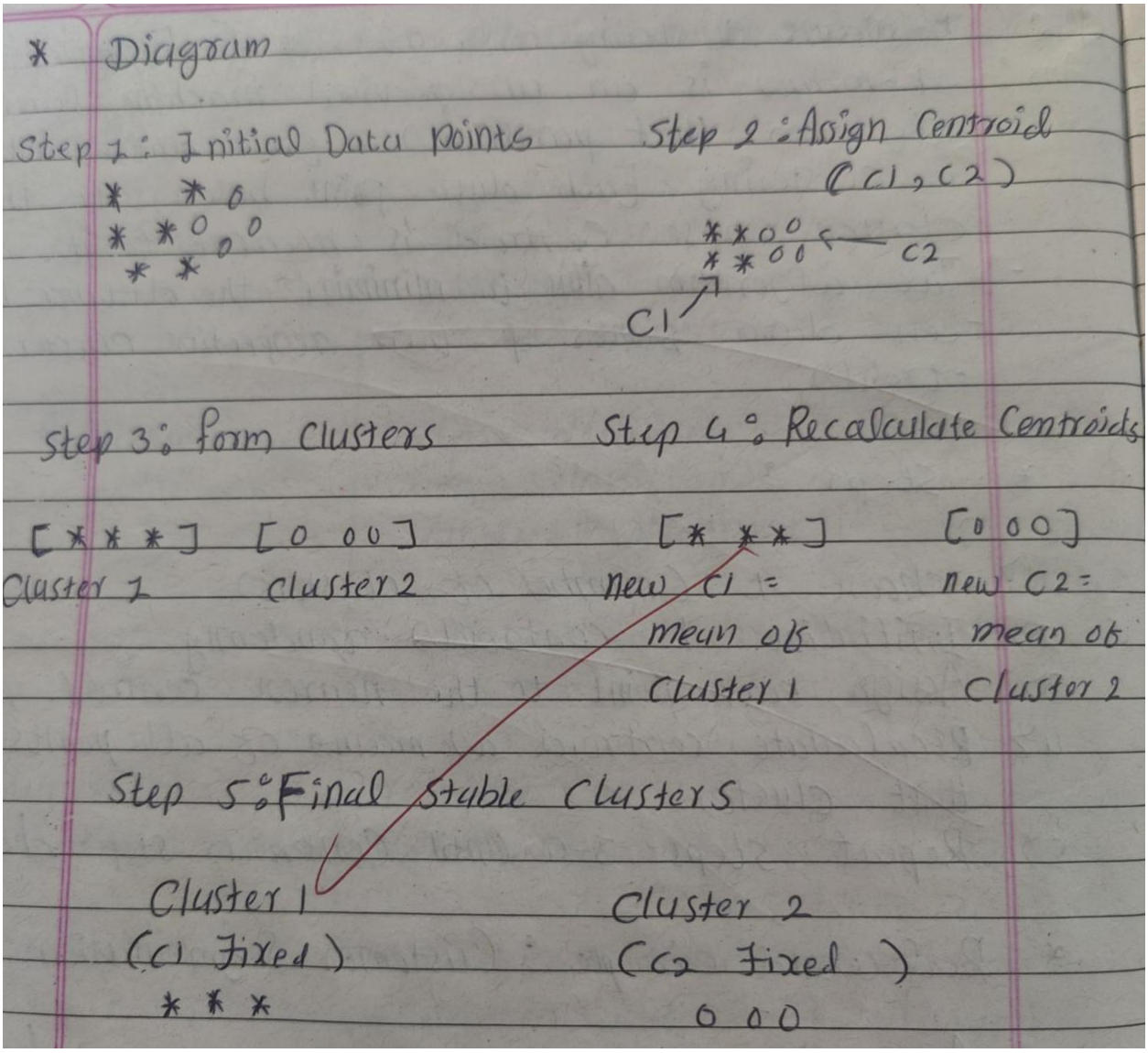
Assign each data point to nearest centroid

Recalculate centroids of clusters

Repeat steps 3 and 4 until no change in clusters

Real-world Example:

In customer segmentation, a company groups customers based on purchasing behavior. Customers with similar buying patterns are placed in the same cluster, helping in targeted marketing.



Q18. Write and explain Apriori Algorithm in detail

Apriori algorithm is an unsupervised learning technique used to find frequent itemsets and generate association rules from a dataset. It is mainly used in market basket analysis to discover relationships between items.

The algorithm works on the principle that if an itemset is frequent, then all of its subsets must also be frequent. This is known as the Apriori property.

Steps of Apriori Algorithm:

Set minimum support and confidence values.

Find all frequent 1-itemsets that satisfy minimum support.

Generate candidate 2-itemsets from frequent 1-itemsets.

Calculate support and keep only those itemsets that satisfy minimum support.

Repeat the process to generate higher-order itemsets until no more frequent itemsets are found.

Generate association rules from frequent itemsets using confidence.

Keep only those rules that satisfy minimum confidence.

Example:

If customers frequently buy bread and butter together, Apriori will generate a rule like:

Bread → Butter

Thus, Apriori helps in identifying strong relationships between items in large datasets.

Q19. Define the following: (1) Support (2) Confidence

Support is the measure of how frequently an itemset appears in the dataset. It shows the proportion of transactions that contain a particular itemset.

Support = (Number of transactions containing the itemset) / (Total number of transactions)

Confidence is the measure of how often a rule is found to be true. It indicates the reliability of the rule.

Confidence = (Number of transactions containing both A and B) / (Number of transactions containing A)

Support helps in identifying frequent itemsets, while confidence helps in evaluating the strength of association rules.

Q20. Define Classification. Illustrate classification learning steps in detail

Classification is a supervised learning technique used to assign input data into predefined categories or classes. It predicts discrete outputs such as Yes/No, Spam/Not Spam, etc.

Steps in Classification Learning:

Data Collection: Gather labeled dataset with input features and corresponding class labels.

Data Preprocessing: Clean the data, handle missing values, and normalize if required.

Feature Selection: Select important features that affect the output.

Model Selection: Choose a suitable classification algorithm like Decision Tree, SVM, or K-NN.

Training: Train the model using the labeled dataset.

Testing: Evaluate the model using test data to check accuracy.

Prediction: Use the trained model to classify new unseen data.